SM Graded Quiz M1

Question 1

What is the characteristic function approach mainly used for in the Heston model?

- Deriving semi-closed form solutions for European option prices
- Calibrating model parameters from historical data
- Simulating stock price paths
- Pricing American options

Question 2

Why must the Heston model satisfy the Feller condition $2\kappa\theta > \sigma^2$?

- To enable closed-form solutions
- To ensure variance stays strictly positive
- To guarantee mean reversion
- To maintain martingale property

Question 3

What happens to the Heston model as the vol-of-vol parameter $\sigma \to 0$?

- The model becomes undefined
- It converges to a jump-diffusion model
- It converges to Black-Scholes with constant volatility $\sqrt{\theta}$
- Variance grows without bound

Question 4

Which parameter set would you recommend for modeling equity index options during normal market conditions?

- $\kappa = 1, \theta = 0.2, \sigma = 1, \rho = 0.5, \nu_0 = 0.2$
- $\kappa = 0.1, \theta = 0.5, \sigma = 2, \rho = 0.8, \nu_0 = 0.5$
- $\kappa = 3, \theta = 0.04, \sigma = 0.3, \rho = -0.7, \nu_0 = 0.04$

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- [x] $\kappa = 10, \theta = 0.01, \sigma = 0.01, \rho = 0, \nu_0 = 0.01$

Question 5

If the correlation ρ changes from 0.25 to -0.5, with $v_0 = 0.1$ and $\theta = 0.1$, what is the new covariance matrix element at position [0,1]?

- 0.75
- -0.5
- -0.25
- [x] 0.5

Question 6

For a deep out-of-the-money put with $S_0 = 100$, $K = 70, r = 0.05, T = 1, v_0 = 0.04, \kappa = 2$, $\theta = 0.04, \sigma = 0.3, \rho = -0.5$, what is the approximate price?

- 0.02
- 5.25
- 2.1
- [x] 0.35

Question 7

Why does the code use `np.sqrt(v[t-1]) * sigma` in the variance equation instead of `sigma * v[t-1]`?

- To prevent variance explosion
- To match the volatility scaling in the stock price equation
- Because the CIR process has volatility proportional to square root of variance
- [x] To ensure faster convergence of the simulation

Question 8

What is the impact of using row indices 0 and 1 in the `cho_matrix` multiplication for S and V paths respectively?

- It defines the drift terms for each process
- It sets the order of the random number generation
- It determines which process gets calculated first

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- [x] It ensures proper correlation structure between the two Brownian motions

Question 9

If $\lambda = 2$ (jumps per year), $\mu_j = -0.05$, and $\delta = 0.1$, what is the value of r_j?

- 0.091
- -0.088
- -0.182
- [x] 0.045

Question 10

Which market phenomenon is the Merton model particularly good at capturing compared to Black-Scholes?

- Mean reversion in volatility
- Interest rate changes
- Volatility clustering over time
- [x] Large sudden price movements and fat tails in return distributions

Question 11

How does the Merton model's implied volatility surface differ from Black-Scholes?

- It reduces implied volatility for all strikes
- It only affects at-the-money options
- It exhibits volatility smiles even for longer maturities
- [x] It produces a flat implied volatility surface

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Question 12

For risk management, which parameter in the Merton model is most critical for tail risk assessment?

- Only the diffusion volatility σ
- The jump parameters λ, μ_j, and δ collectively
- The risk-free rate r
- The initial stock price S_0

Question 13

In the code, why is `SM[t] = np.maximum(SM[t], 0.00001)` used after each time step?

- To implement a price floor for the option
- To ensure jumps are always positive
- To prevent numerical issues from negative or zero prices
- To cap the maximum loss

Question 14

With `lamb=0.75`, `mu=-0.6`, `delta=0.25`, what is the value of `rj` used in the drift adjustment?

- -0.45
- -0.325
- 0.391
- -0.522

Question 15

Which scenario would show the largest difference between Merton and Black-Scholes prices?

- Short maturity options with high jump intensity and large jump volatility
- Long maturity at-the-money options
- Deep in-the-money calls with low volatility
- Options with zero interest rate

Question 16

What is the impact of setting `delta=0` while keeping `lamb>0` in the implementation?

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- All jumps would have the same deterministic size $\exp(\mu) - 1$
- No jumps would occur
- Jump sizes would be zero
- The model reduces to Black-Scholes

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